

Module - IV: Partial Differential Equations...

Partial Differential Equation

Any equation containing one or more partial derivative of an unknown function of two or more independent variables is known as partial differential equation - (PDE).

Example:

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = z + \sin y \quad \text{--- (1)}$$

$$\left(\frac{\partial z}{\partial x}\right)^2 + \frac{\partial^3 z}{\partial y^3} = 2x \frac{\partial z}{\partial x} \quad \text{--- (2)}$$

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = xyz \quad \text{--- (3)}$$

$$\frac{\partial^2 z}{\partial x^2} = \left(1 + \frac{\partial z}{\partial y}\right)^{1/2} \quad \text{--- (4)}$$

Order of P.D.E. :

order of the

The highest partial derivative occurring in the P.D.E. is known as the order of P.D.E.

Order of eqⁿ. (1) is 1, eqⁿ. (2) is 3, eqⁿ. (3) is 1 and eqⁿ. (4) is 2.

Degree of P.D.E. :

order

The degree of the highest derivative is known as the degree of P.D.E.

Degree of (1) is 1,

(2) is 1,

(3) is 1,

(4) is 2

$$\rightarrow \left(\frac{\partial^2 z}{\partial x^2} \right)^2 = 1 + \frac{\partial z}{\partial y}$$

Linear & Non-linear Partial Differential Equation

If the dependent variable and its partial derivative occurs only in the first degree and not multiplied together, then the equation is known as linear P.D.E., otherwise non-linear P.D.E.

Formation of Partial Differential Equation

Any equation containing arbitrary constant then the replacement of this arbitrary constant by partial derivative is known as the formation of P.D.E.

Question

1. Find a partial differential equation by eliminating 'a' & 'b' from $z = ax + by + a^2 + b^2$.

Solⁿ:

The given equation is,
 $z = ax + by + a^2 + b^2$ — (1)

Partially differentiating eqⁿ. (1) w.r.to x ,
 $\frac{\partial z}{\partial x} = a$

Again, partially differentiating eqⁿ. (1) w.r.to y ,
 $\frac{\partial z}{\partial y} = b$

Putting a & b in eqⁿ. (1), we get,

$$z = x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} + \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2,$$

which is the required P.D.E.

2. Obtain P.D.E. of $z = A e^{pt} \sin(px)$ by eliminating A & p .

Solⁿ:

The given eqⁿ. is,

$$z = A e^{pt} \sin(px) \quad \text{--- (1)}$$

Partially differentiating eqⁿ. (1) w.r.to x ,

$$\frac{\partial z}{\partial x} = A p e^{pt} \cos px$$

$$\Rightarrow \frac{\partial^2 z}{\partial x^2} = -A p^2 e^{pt} \sin px$$

$$\Rightarrow \frac{\partial^2 z}{\partial x^2} = -p^2 z \quad \text{--- (2)}$$

Partially differentiating eqⁿ. (1) w.r.to t ,

$$\frac{\partial z}{\partial t} = A p e^{pt} \sin px$$

$$\Rightarrow \frac{\partial z}{\partial t} = p z \quad \text{--- (3)}$$

From eqⁿ. (2) & eqⁿ. (3),

$$-\frac{1}{z} \frac{\partial^2 z}{\partial x^2} = \left(\frac{1}{z} \frac{\partial z}{\partial t} \right)^2$$

$$\Rightarrow -\frac{1}{z} \frac{\partial^2 z}{\partial x^2} = \frac{1}{z^2} \left(\frac{\partial z}{\partial t} \right)^2$$

$$\Rightarrow z \frac{\partial^2 z}{\partial x^2} + \left(\frac{\partial z}{\partial t} \right)^2 = 0, \text{ is the required P.D.E.}$$

3. Find P.D.E. of $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ by eliminating a, b & c .

Solⁿ:

The given eqⁿ. is,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \text{--- (1)}$$

Partially differentiating w.r.to x ,

$$\frac{2x}{a^2} + \frac{2z}{c^2} \frac{\partial z}{\partial x} = 0$$

$$\Rightarrow \frac{x}{a^2} + \frac{z}{c^2} \frac{\partial z}{\partial x} = 0 \quad \text{--- (2)}$$

Partially differentiating w.r.to y ,

$$\frac{2y}{b^2} + \frac{2z}{c^2} \frac{\partial z}{\partial y} = 0$$

$$\Rightarrow \frac{y}{b^2} + \frac{z}{c^2} \frac{\partial z}{\partial y} = 0 \quad \text{--- (3)}$$

Again,

Partially differentiating eqⁿ. (2) w.r.to x ,

$$\frac{1}{a^2} + \frac{1}{c^2} \left[z \frac{\partial^2 z}{\partial x^2} + \left(\frac{\partial z}{\partial x} \right)^2 \right] = 0 \quad \text{--- (4)}$$

Partially differentiating eqⁿ. (3) w.r.to y ,

$$\frac{1}{b^2} + \frac{1}{c^2} \left[z \frac{\partial^2 z}{\partial y^2} + \left(\frac{\partial z}{\partial y} \right)^2 \right] = 0 \quad \text{--- (5)}$$

From eqⁿ. (2)

$$\frac{x}{a^2} = -\frac{z}{c^2} \cdot \frac{\partial z}{\partial x}$$
$$\Rightarrow \frac{1}{a^2} = -\frac{z}{xc^2} \cdot \frac{\partial z}{\partial x}$$

Putting this $1/a^2$ in eqⁿ. (4),

$$\frac{-z}{xc^2} \cdot \frac{\partial z}{\partial x} + \frac{1}{c^2} \left[z \frac{\partial^2 z}{\partial x^2} + \left(\frac{\partial z}{\partial x} \right)^2 \right] = 0$$
$$\Rightarrow -\frac{z}{x} \frac{\partial z}{\partial x} + z \frac{\partial^2 z}{\partial x^2} + \left(\frac{\partial z}{\partial x} \right)^2 = 0$$
$$\Rightarrow -z \frac{\partial z}{\partial x} + zx \frac{\partial^2 z}{\partial x^2} + x \left(\frac{\partial z}{\partial x} \right)^2 = 0,$$

which is the required P.D.E.

Formation of P.D.E. by Eliminating Arbitrary Function

$$\phi(u, v) = 0 \text{ ————— (1)}$$

Then u & v are function of x, y & z . Then P.D.E. of eqⁿ. (1) is formed by eliminating arbitrary function.

1. Form a P.D.E. by eliminating the arbitrary constant ϕ from $\phi(x+y+z, x^2+y^2-z^2) = 0$. What is the order of the differential equation.

Solⁿ:

The given function is,

$$\phi(x+y+z, x^2+y^2-z^2) = 0 \text{ ————— (1)}$$

$$\text{Here, } u = x+y+z$$

$$\& \ v = x^2+y^2-z^2$$

Partially Differentiating eqⁿ. (1) w.r.to x ,

$$\frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right) = 0 \text{ — (2)}$$

$$\Rightarrow \frac{\partial \phi}{\partial u} (1+p) + \frac{\partial \phi}{\partial v} (2x - 2zp)$$

$$\Rightarrow \frac{\partial \phi}{\partial u} (1+p) = -2 \frac{\partial \phi}{\partial v} (x-zp)$$

$$\Rightarrow \frac{\frac{\partial \phi}{\partial u}}{\frac{\partial \phi}{\partial v}} = \frac{-2(x-zp)}{(1+p)} \quad \text{--- (3)}$$

Partially differentiating eqⁿ. (1) w.r. to y ,

$$\frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \right) = 0 \quad \text{--- (4)}$$

$$\Rightarrow \frac{\partial \phi}{\partial u} (1+q) + \frac{\partial \phi}{\partial v} (2y-2zq) = 0$$

$$\Rightarrow \frac{\frac{\partial \phi}{\partial u}}{\frac{\partial \phi}{\partial v}} = \frac{-2(y-zq)}{1+q} \quad \text{--- (5)}$$

From eqⁿ. (3) & (5),

$$\frac{-2(x-zp)}{1+p} = \frac{-2(y-zq)}{1+q}$$

$$\Rightarrow x + qx - zp - zpq = y - zq + py - zpq$$

$$\Rightarrow (x-y) + qx - py - zp + zq = 0$$

$$\Rightarrow (x-y) + q(x+z) - p(y+z) = 0$$

which is the required P.D.E.

2. Eliminate the arbitrary function f & F from $y = f(x - at) + F(x + at)$.

Solⁿ:

The given is,

$$y = f(x - at) + F(x + at) \quad \text{--- (1)}$$

Then,

$$\frac{\partial y}{\partial x} = f'(x - at) + F'(x + at)$$

$$\frac{\partial^2 y}{\partial x^2} = f''(x - at) + F''(x + at) \quad \text{--- (2)}$$

Again,

$$\frac{\partial y}{\partial t} = f'(x - at)(-a) + F'(x + at)(a)$$

$$\frac{\partial^2 y}{\partial t^2} = f''(x - at)(-a)^2 + F''(x + at)(a)^2 \quad \text{--- (3)}$$

$$\Rightarrow \frac{\partial^2 y}{\partial t^2} = a^2 \{ f''(x - at) + F''(x + at) \}$$

$$\Rightarrow \frac{\partial^2 y}{\partial t^2} = a^2 \frac{\partial^2 y}{\partial x^2}$$

Solution of P.D.E.

If any equation of type $Pp + Qq = R$ is given then its solution is $\Phi(u, v) = 0$.

Where, P & Q are function of x, y & z . and u & v are also function of x, y & z .

This eqⁿ. is known as Lagrange's Equation.

Question

1. Solve $\left(\frac{y^2 z}{x}\right)p + xz \cdot q = y^2$

Solⁿ:

The given eqⁿ. is,

$$\left(\frac{y^2 z}{x}\right)p + xz \cdot q = y^2 \quad \text{--- (1)}$$

which is a lagrange's equation.

$$\text{i.e. } P = \frac{y^2 z}{x}, Q = xz \text{ \& } R = y^2$$

The lagrange's auxiliary equation is,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{\frac{y^2 z}{x}} = \frac{dy}{xz} = \frac{dz}{y^2} \quad \text{--- (2)}$$

Taking first two part of eqⁿ. (2),

$$\frac{dx}{\frac{y^2 z}{x}} = \frac{dy}{xz}$$

$$\Rightarrow x^2 dx = y^2 dy$$

$$\Rightarrow x^2 dx - y^2 dy = 0$$

Integrating, we get,

$$\frac{x^3}{3} - \frac{y^3}{3} = C_1$$

$$\Rightarrow x^3 - y^3 = C_1 \quad \text{--- (3)}$$

Now, taking first & last part eqⁿ. (2),

$$\frac{dx}{\frac{y^2 z}{x}} = \frac{dz}{y^2}$$

$$\Rightarrow x dx = z dz$$

$$\Rightarrow x dx - z dz = 0$$

$$\Rightarrow \frac{x^2}{2} - \frac{z^2}{2} = C_2$$

$$\Rightarrow x^2 - z^2 = C_2 \quad \text{--- (4)}$$

From eqⁿ. (3) & (4),

$$\Phi(x^3 - y^3, x^2 - z^2) = 0$$

which is the required solution.

2. Solve: $p + 3q = 5z + \tan(y - 3x)$

Solⁿ:

The given P.D.E is,

$$p + 3q = 5z + \tan(y - 3x) \quad \text{--- (1)}$$

The lagrange's auxiliary equatis is,

$$\frac{dx}{1} = \frac{dy}{3} = \frac{dz}{5z + \tan(y - 3x)} \quad \text{--- (2)}$$

Taking first & second part of eqⁿ. (2), we get,

$$\begin{aligned} 3dx &= dy \\ \Rightarrow 3x - y &= C_1 \\ \Rightarrow y - 3x &= C_1 \quad \text{--- (3)} \end{aligned}$$

Taking first & third part of eqⁿ. (3),

$$\frac{dx}{1} = \frac{dz}{5z + \tan(y - 3x)} \quad \text{--- (4)}$$

$$\Rightarrow dx = \frac{dz}{5z + \tan C_1} \quad [\text{From eqⁿ. (3)}]$$

$$\Rightarrow x = \frac{1}{5} \log(5z + \tan C_1) + C_2$$

$$\Rightarrow 5x - \log(5z + \tan C_1) = C_2 \quad \text{--- (5)}$$

$$\Rightarrow 5x - \log(5z + \tan(y - 3x)) = C_2 \quad \text{--- (5)}$$

From eqⁿ. (3) & (5),

$$\Phi(y-3x, 5x - \log[5z + \tan(y-3x)]) = 0$$

This is the required solution of given P.D.E.

Homogeneous Linear Differential Equation with Constant Coefficient

If any equation is in form of auxiliary eqⁿ.
i.e.

$$\{A_0 D^n + A_1 D^{n-1} D' + \dots + A_n D'^n\} z = f(x, y)$$

Case-I: If ~~m~~ $m_1, m_2, m_3, \dots$

$$C.F. = \Phi_1(y + m_1 x) + \Phi_2(y + m_2 x) + \dots$$

Case-II: If $m_1 = m_2, m_3, m_4, \dots$

$$C.F. = \Phi_1(y + m_1 x) + x \Phi_2(y + m_2 x) + \dots$$

Case-III: Corresponding to a non-repeated factor D on LHS of (1) then

$$C.F. = \Phi(y)$$

Case-IV: Repeated...

$$C.F. = \Phi(y) + x \Phi_2(y) + \dots$$

Important Notation

$$p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y}$$

$$\frac{\partial}{\partial x} \equiv D$$

$$r = \frac{\partial^2 z}{\partial x^2}, \quad s = \frac{\partial^2 z}{\partial x \partial y}$$

$$\frac{\partial}{\partial y} \equiv D'$$

$$t = \frac{\partial^2 z}{\partial y^2}$$

Questions

1. $\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = 0$, find the solution of the given P.D.E.

Solⁿ:

The given P.D.E is,

$$\frac{\partial^2 z}{\partial x^2} - \frac{\partial^2 z}{\partial y^2} = 0$$

Replacing, $\frac{\partial}{\partial x} \equiv D$ & $\frac{\partial}{\partial y} \equiv D'$,

$$D^2 - (D')^2 = 0$$

$$\Rightarrow (D - D')(D + D') = 0$$

$$\Rightarrow (m - 1)(m + 1) = 0 \quad [D = m \text{ \& \; } D' = 1]$$

$$\Rightarrow m = 1, m = -1$$

$$C.F. = \phi_1(y+x) + \phi_2(y-x)$$

This is the required solution.

2. Solve: $2x + 5s + 2t = 0$

Solⁿ:

The given eqⁿ is,

$$2x + 5s + 2t = 0$$

It can be written as,

$$2 \frac{\partial^2 z}{\partial x^2} + 5 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial y^2} = 0$$

Now, $\frac{\partial}{\partial x} \equiv D$ & $\frac{\partial}{\partial y} \equiv D'$

$$2D^2 + 5D.D' + 2D'^2 = 0$$

$$\Rightarrow 2m^2 + 5m + 2 = 0 \quad [D = m \text{ \& } D' = 1]$$

$$\Rightarrow 2m^2 + 4m + m + 2 = 0$$

$$\Rightarrow 2m(m+2) + 1(m+2) = 0$$

$$\Rightarrow (2m+1)(m+2) = 0$$

$$\Rightarrow m = -1/2, m = 2$$

$$C.F. = \phi_1(y - 1/2 x) + \phi_2(y + 2x)$$

This is the required solution.

3. Solve: $(D^3 - 4D^2.D' + 4DD'^2)z = 0$

Solⁿ:

The given eqⁿ. is,

$$(D^3 - 4D^2.D' + 4DD'^2)z = 0$$

$$\Rightarrow m^3 - 4m^2 + 4m = 0 \quad [\because D=m \text{ \& } D'=1]$$

$$\Rightarrow m(m^2 - 4m + 4) = 0$$

$$\Rightarrow m(m-2)^2 = 0$$

$$\Rightarrow m = 0, 2, 2$$

$$C.F. = \phi_1(y) + \phi_2(y+2x) + x\phi_3(y+2x).$$

This is ~~the~~ the required solution.

4. Solve: $(D^4 - 2D^3.D' + 2D.D'^3 - D'^4)z = 0$

Solⁿ:

The given eqⁿ. can be written as,

$$m^4 - 2m^3 + 2m - 1 = 0$$

$$\Rightarrow (m^4 - 1) - (2m^3 - 2m) = 0$$

$$\Rightarrow (m^2 + 1)(m^2 - 1) - 2m(m^2 - 1) = 0$$

$$\Rightarrow [(m^2 + 1) - 2m](m^2 - 1) = 0$$

$$\Rightarrow (m^2 - 1)(m - 1)^2 = 0$$

$$\Rightarrow (m + 1)(m - 1)(m - 1)(m - 1) = 0$$

$$\Rightarrow m = -1, 1, 1, 1$$

$$\therefore C.F. = \phi_1(y-x) + \phi_2(y+x) + x\phi_3(y+x) + x^2\phi_4(y+x),$$

is the required solution.

Particular Integral

$$F(D, D')z = f(x, y)$$

1. Solve: $(D^2 + 3DD' + 2D'^2)z = x + y$

Solⁿ:

The given eqⁿ. is,

$$(D^2 + 3DD' + 2D'^2)z = x + y$$

The A.E. is,

$$D^2 + 3DD' + 2D'^2 = 0$$

$$\Rightarrow m^2 + 3m + 2 = 0 \quad [\because D = m \text{ \& } D' = 1]$$

$$\Rightarrow m^2 + 2m + m + 2 = 0$$

$$\Rightarrow m(m+2) + 1(m+2) = 0$$

$$\Rightarrow (m+1)(m+2) = 0$$

$$\Rightarrow m = -1, -2$$

$$C.F. = \Phi_1(y-x) + \Phi_2(y-2x)$$

$$f(x, y) = x + y = v$$

Comparing it with $ax + by$, $a = 1$ & $b = 1$
 $\Rightarrow D = 1$ & $D' = 1$

$$P.I. = \frac{x+y}{D^2 + 3DD' + 2D'^2}$$

$$= \frac{1}{1+3+2} \iint v \, dv$$

$$= \frac{1}{6} \cdot \frac{v^3}{6}$$

$$= \frac{(x+y)^3}{36}$$

∴ The solution is,

$$z = \phi_1(y-x) + \phi_2(y-2x) + \frac{(x+y)^3}{36}$$

2. Solve:

$$x + y - 2z = (2x + y)^{1/2}$$

Solⁿ:

The given eqⁿ. can be written as,

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 2 \frac{\partial^2 z}{\partial y^2} = (2x + y)^{1/2}$$

$$\frac{\partial}{\partial x} \equiv D \quad \& \quad \frac{\partial}{\partial y} \equiv D'$$

The A.E. is,

$$D^2 + DD' - 2D'^2 = 0$$

$$\Rightarrow m^2 + m - 2 = 0 \quad [\because D = m \quad \& \quad D' = 1]$$

$$\Rightarrow m^2 + (2-1)m - 2 = 0$$

$$\Rightarrow m^2 + 2m - m - 2 = 0$$

$$\Rightarrow m(m+2) - 1(m+2) = 0$$

$$\Rightarrow (m-1)(m+2) = 0$$

$$\Rightarrow m = 1, -2$$

$$C.F. = \phi_1(y+x) + \phi_2(y-2x)$$

$$f(x, y) = (2x + y)^{1/2} = v^{1/2}$$

Comparing it with $ax + by$, we get,

$$\& \quad a=2, \quad b=1 \\ D=2, \quad D'=1$$

$$P.I. = \frac{(2x+y)^{1/2}}{D^2 + DD' - 2D'^2}$$

$$= \frac{1}{2^2 + 2 \times 1 - 2 \times 1^2} \iint v^{1/2} dv$$

$$= \frac{1}{4} \cdot \frac{v^{5/2}}{\frac{3}{2} \times \frac{5}{2}}$$

$$= \frac{v^{5/2}}{15}$$

$$= \frac{(2x+y)^{5/2}}{15}$$

Hence, the solution of given P.D.E. is,

$$z = \Phi_1(y+x) + \Phi_2(y-2x) + \frac{1}{15} (2x+y)^{5/2}$$

Solution of P.D.E. when $f(x, y) = e^{ax+by}$

1. Solve: $(D^2 + 2DD' + D'^2)z = e^{2x+3y}$

Solⁿ:

The given P.D.E. is,

$$(D^2 + 2DD' + D'^2)z = e^{2x+3y}$$

The A.E. is,

$$D^2 + 2DD' + D'^2 = 0$$

$$\Rightarrow m^2 + 2m + 1 = 0 \quad [D = m \text{ \& } D' = 1]$$

$$\Rightarrow (m+1)^2 = 0$$

$$\Rightarrow m = -1, -1$$

$$C.F. = \phi_1(y-x) + x\phi_2(y-x)$$

$$\text{Let, } 2x + 3y = v \Rightarrow D = 2 \text{ \& } D' = 3$$

$$P.I. = \frac{e^{2x+3y}}{D^2 + 2DD' + D'^2}$$

$$= \frac{1}{2^2 + 2 \times 2 \times 3 + 3^2} \iint e^v dv$$

$$= \frac{1}{25} e^v$$

$$= \frac{1}{25} e^{2x+3y}$$

$\therefore$ The solⁿ is,

$$z = \phi_1(y-x) + x\phi_2(y-x) + \frac{1}{25} e^{2x+3y}$$

2. Solve: $(D^2 - 2DD' + D'^2)z = e^{x+2y}$

Solⁿ:

The A.E. of given P.D.E. is,

$$D^2 - 2DD' + D'^2 = 0$$

$$\Rightarrow m^2 - 2m + 1 = 0 \quad [D=m \text{ \& } D'=1]$$

$$\Rightarrow (m-1)^2 = 0$$

$$\Rightarrow m=1, 1$$

$$C.F. = \phi_1(y+x) + x\phi_2(y+x)$$

$$\text{Let, } x+2y = v \Rightarrow D=1 \text{ \& } D'=2$$

$$P.I. = \frac{e^{x+2y}}{D^2 - 2DD' + D'^2}$$

$$= \frac{1}{1^2 - 2 \times 1 \times 2 + 2^2} \iint e^v dv$$

$$= e^v$$

$$= e^{x+2y}$$

$\therefore$ The solution is,

$$z = \phi_1(y+x) + x\phi_2(y+x) + e^{x+2y}$$

Solution of P.D.E when $f(x,y) = \sin(ax+by)$ or $f(x,y) = \cos(ax+by)$

1. Solve: $x - 2s + t = \sin(2x + 3y)$

Solⁿ:

The given P.D.E. is,

$$\begin{aligned} x - 2s + t &= \sin(2x + 3y) \\ \Rightarrow \frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} &= \sin(2x + 3y) \\ \Rightarrow (D^2 - 2DD' + D'^2)z &= \sin(2x + 3y) \end{aligned}$$

The A.E. is,

$$\begin{aligned} D^2 - 2DD' + D'^2 &= 0 \\ \Rightarrow m^2 - 2m + 1 &= 0 \quad [D=m \text{ \& } D'=1] \\ \Rightarrow (m-1)^2 &= 0 \\ \Rightarrow m &= 1, 1 \end{aligned}$$

$$C.F. = \phi_1(y+x) + x \phi_2(y+x)$$

$$\text{Let, } 2x + 3y = v \Rightarrow D = 2 \text{ \& } D' = 3$$

$$P.I. = \frac{\sin(2x + 3y)}{D^2 - 2DD' + D'^2}$$

$$= \frac{1}{2^2 - 2 \times 2 \times 3 + 3^2} \iint \sin v \, dv$$

$$= \frac{+1}{10} (-\sin v)$$

$$\therefore \text{P.I.} = -\frac{1}{10} \sin(2x+3y) = -\sin(2x+3y)$$

Hence, the solution is,

$$z = \phi_1(y+ix) + x\phi_2(y+ix) - \sin(2x+3y)$$

2. Solve: $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \cos mx \cdot \sin ny$

Solⁿ:

The given P.D.E. is,

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = \cos mx \cdot \sin ny$$

$$\Rightarrow (D^2 + D'^2)z = \cos mx \cdot \sin ny$$

$$= \frac{1}{2} [\sin(mx+ny) + \sin(mx-ny)]$$

The A.E. is,

$$D^2 + D'^2 = 0$$

$$\Rightarrow m^2 + 1 = 0 \quad [D = m \text{ \& } D' = 1]$$

$$\Rightarrow m^2 = -1$$

$$\Rightarrow m = \pm i$$

$$\text{C.F.} = \phi_1(y+ix) + \phi_2(y-ix)$$

$$\text{P.I.} = \frac{1}{D^2 + D'^2} \left(\frac{1}{2} [\sin(mx+ny) + \sin(mx-ny)] \right)$$

$$= \frac{1}{2} \cdot \frac{1}{D^2 + D'^2} \sin(mx+ny) + \frac{1}{2} \cdot \frac{1}{D^2 + D'^2} \sin(mx-ny)$$

Let, $mx + ny = v_1 \Rightarrow D = m \text{ \& } D' = n$
 Also, $mx - ny = v_2 \Rightarrow D = m \text{ \& } D' = -n$

$$P.I. = \frac{1}{2} \cdot \frac{1}{m^2 + n^2} \iint \sin v_1 \, dv_1 + \frac{1}{2} \cdot \frac{1}{m^2 + (-n)^2} \iint \sin v_2 \, dv_2$$

$$= \frac{1}{2(m^2 + n^2)} \left[-\sin v_1 + (-\sin v_2) \right]$$

$$= \frac{-1}{2(m^2 + n^2)} [\sin(mx + ny) + \sin(mx - ny)]$$

$$= \frac{-1}{2(m^2 + n^2)} \cdot \cos mx \cdot \sin ny$$

Hence, the solution of given P.D.E is,

$$z = \phi_1(y + ix) + \phi_2(y - ix) - \frac{\cos mx \cdot \sin ny}{m^2 + n^2}$$

Solution of P.D.E. when $\frac{1}{(bD - aD')^n} \cdot \phi(ax + by)$

$$= \frac{x^n}{b^n n!} \phi(ax + by)$$

1. Solve: $4x - 4s + t = 16 \log(x + 2y)$

Solⁿ:

The given P.D.E. is,

$$4x - 4s + t = 16 \log(x + 2y)$$

$$\Rightarrow 4 \frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = 16 \log(x + 2y)$$

$$\Rightarrow (4D^2 - 4DD' + D'^2)z = 16 \log(x + 2y)$$

The A.E. is,

$$4D^2 - 4DD' + D'^2 = 0$$

$$\Rightarrow 4m^2 - 4m + 1 = 0 \quad [D = m \text{ \& } D' = 1]$$

$$\Rightarrow (2m - 1)^2 = 0$$

$$\Rightarrow m = 1/2, 1/2$$

$$C.F. = \phi_1(y + 1/2 x) + x \phi_2(y + 1/2 x)$$

$$\Rightarrow C.F. = \phi_1(2y + x) + x \phi_2(2y + x)$$

$$P.I. = \frac{16 \log(x + 2y)}{4D^2 - 4DD' + D'^2}$$

$$= \frac{1}{(2D - D')^2} \cdot 16 \log(x + 2y)$$

$$= \frac{x^2}{2^2 2!} \cdot 16 \log(x+2y) \quad [\text{Using Formula}]$$

$$= 2x^2 \log(x+2y)$$

Hence, the solution of given P.D.E. is,

$$z = \phi_1(2y+x) + x\phi(2y+x) + 2x^2 \log(x+2y).$$

Solution of P.D.E. when the function $f(x,y)=x^n y^n$

1. Solve: $x + (a+b)s + ab.t = xy$

Solⁿ:

The given P.D.E. is,

$$x + (a+b)s + ab.t = xy$$
$$\Rightarrow \frac{\partial^2 z}{\partial x^2} + (a+b) \frac{\partial^2 z}{\partial x \partial y} + ab \frac{\partial^2 z}{\partial y^2} = xy$$

$$\Rightarrow D^2 + (a+b)DD' + ab D'^2 = xy$$

The A.E. is,

$$D^2 + (a+b)DD' + ab D'^2 = 0$$

$$\Rightarrow D^2 + aDD' + bDD' + ab D'^2 = 0$$

$$\Rightarrow D(D+aD') + bD'(D+aD') = 0$$

$$\Rightarrow (D+bD')(D+aD') = 0$$

$$\Rightarrow (m+b)(m+a) = 0 \quad [D=m \text{ \& } D'=1]$$

$$\Rightarrow m = -b, -a$$

$$C.F. = \phi_1(y-bx) + \phi_2(y-ax)$$

$$P.I. = \frac{1}{D^2 + (a+b)DD' + ab D'^2} \cdot xy$$

$$= \frac{1}{D^2 \left[1 + (a+b) \frac{D'}{D} + ab \frac{D'^2}{D^2} \right]} \cdot xy$$

$$= \frac{1}{D^2} \left[1 + (a+b) \frac{D'}{D} + ab \frac{D'^2}{D^2} \right]^{-1} \cdot xy$$

$$= \frac{1}{D^2} \left[1 - (a+b) \frac{D'}{D} - ab \frac{D'^2}{D^2} + \dots \right] xy$$

$$= \frac{1}{D^2} \left[xy - (a+b) \frac{x^2}{2} \right]$$

$$= \frac{x^3 y}{6} - \frac{(a+b)x^4}{24}$$

Hence, the solution is,

$$z = \Phi_1(y-bx) + \Phi_2(y-ax) + \frac{x^3 y}{6} - \frac{(a+b)x^4}{24}$$

2. Solve: $(D^2 - 6DD' + 9D'^2)z = 12x^2 + 36xy$

Solⁿ:

The A.E. of given P.D.E. is,

$$D^2 - 6DD' + 9D'^2 = 0$$

$$\Rightarrow m^2 - 6m + 9 = 0 \quad [D = m \text{ \& } D' = 1]$$

$$\Rightarrow m^2 - (3+3)m + 9 = 0$$

$$\Rightarrow m^2 - 3m - 3m + 9 = 0$$

$$\Rightarrow m(m-3) - 3(m-3) = 0$$

$$\Rightarrow (m-3)(m-3) = 0$$

$$\Rightarrow m = 3, 3$$

$$C.F. = \phi_1(y+3x) + x\phi_2(y+3x)$$

$$P.I. = \frac{1}{D^2 - 6DD' + 9D'^2} \cdot (12x^2 + 36xy)$$

$$= \frac{1}{D^2 \left[1 - \frac{6D'}{D} + \frac{9D'^2}{D^2} \right]} \cdot (12x^2 + 36xy)$$

$$= \frac{1}{D^2} \left[1 - \frac{6D'}{D} + \frac{9D'^2}{D^2} \right]^{-1} (12x^2 + 36xy)$$

$$= \frac{1}{D^2} \left[1 + \frac{6D'}{D} + \frac{9D'^2}{D^2} + \dots \right] (12x^2 + 36xy)$$

$$= \frac{1}{D^2} \left[12x^2 + 36xy + \frac{216x^2}{2} + 0 \right]$$

$$= \frac{1}{D^2} (120x^2 + 36xy) = 10x^4 + 6x^3y$$

$$\therefore z = \phi_1(y+3x) + x\phi_2(y+3x) + 10x^4 + 6x^3y.$$

Solution of P.D.E. when the function $f(x,y) = ve^x$

Q.1. Solve: $(D^2 - DD' - 2D'^2)z = (y-1)e^x$

Solⁿ:

The given P.D.E. is,

$$(D^2 - DD' - 2D'^2)z = (y-1)e^x$$

The A.E. is,

$$D^2 - DD' - 2D'^2 = 0$$

$$\Rightarrow m^2 - m - 2 = 0$$

$$\Rightarrow m^2 - 2m + m - 2 = 0$$

$$\Rightarrow m(m-2) + 1(m-2) = 0$$

$$\Rightarrow (m+1)(m-2) = 0$$

$$\Rightarrow m = -1, 2$$

$$C.F. = \phi_1(y-x) + \phi_2(y+2x)$$

$$P.I. = \frac{1}{D^2 - DD' - 2D'^2} \cdot (y-1)e^x$$

$$= \frac{1}{(D+D')(D-2D')} \cdot (y-1)e^x$$

Putting, $y+2x = C_1 \Rightarrow y = C_1 - 2x$

$$P.I. = \frac{1}{(D+D')} \int \underset{I}{(C_1 - 2x - 1)} \underset{II}{e^x} dx$$

$$P.I. = \frac{1}{(D+D')} \left\{ (C_1 - 2x - 1)e^x - \int -2e^x dx \right\}$$

$$= \frac{1}{(D+D')} \left\{ (C_1 - 2x - 1)e^x + 2e^x \right\}$$

$$= \frac{1}{(D+D')} \left\{ (y - 1)e^x + 2e^x \right\}$$

Again, putting, $y - x = C_2 \Rightarrow y = C_2 + x$

$$P.I. = \int \left\{ (C_2 + x - 1)e^x + 2e^x \right\} dx$$

$$= \int (C_2 + x - 1 + 2)e^x dx$$

$$= \int \underbrace{(C_2 + x + 1)}_I e^x dx$$

$$= (C_2 + x + 1)e^x - \int e^x dx$$

$$= (C_2 + x + 1)e^x - e^x$$

$$= (C_2 + x)e^x$$

$$= (y - x + x)e^x = ye^x$$

Hence, the solution is,

$$z = \Phi_1(y - x) + \Phi_2(y + 2x) + ~~(C_2 + x)e^x~~ ye^x.$$

2. Solve: $(D^2 + DD' - 6D'^2)z = y \sin x$

Solⁿ:

The given P.D.E. is,

$$(D^2 + DD' - 6D'^2)z = y \sin x$$

The A.E. is,

$$D^2 + DD' - 6D'^2 = 0$$

$$\Rightarrow m^2 + m - 6 = 0$$

$$\Rightarrow m^2 + 3m - 2m - 6 = 0$$

$$\Rightarrow m(m+3) - 2(m+3) = 0$$

$$\Rightarrow (m-2)(m+3) = 0$$

$$\Rightarrow m = 2, -3$$

$$C.F. = \phi_1(y+2x) + \phi_2(y-3x)$$

$$P.I. = \frac{1}{D^2 + DD' - 6D'^2} \cdot y \sin x$$

$$= \frac{1}{(D-2D')(D+3D')} \cdot y \sin x$$

Putting, $y-3x = C_1 \Rightarrow y = C_1 + 3x$

$$P.I. = \frac{1}{(D-2D')} \int \underset{I}{(C_1 + 3x)} \underset{II}{\sin x} dx$$

$$= \frac{1}{(D-2D')} \left[(C_1 + 3x)(-\cos x) - \int 3(-\cos x) dx \right]$$

$$= \frac{1}{(D-2D')} [-y \cos x + 3 \sin x]$$

Again, putting, $y+2x = C_2 \Rightarrow y = C_2 - 2x$

$$P.I. = \int (C_2 - 2x) \cos x + 3 \sin x \, dx$$

$$= - \left[(C_2 - 2x) \sin x - \int (-2) \sin x \, dx \right] - 3 \cos x$$

$$= - \left[(C_2 - 2x) \sin x - 2 \cos x \right] - 3 \cos x$$

$$= -y \sin x + 2 \cos x - 3 \cos x$$

$$= -y \sin x - \cos x$$

Hence, the solution of given P.D.E. is,

$$z = \Phi_1(y+2x) + \Phi_2(y-3x) - y \sin x - \cos x.$$

3. Solve: $(D^2 + DD' - 6D'^2)z = y \cos x$

Solⁿ:

The A.E. of given P.D.E. is,

$$D^2 + DD' - 6D'^2 = 0$$

$$\Rightarrow m^2 + m - 6 = 0 \quad [D = m \text{ \& } D' = 1]$$

$$\Rightarrow m^2 + (3-2)m - 6 = 0$$

$$\Rightarrow m^2 + 3m - 2m - 6 = 0$$

$$\Rightarrow m(m+3) - 2(m+3) = 0$$

$$\Rightarrow (m-2)(m+3) = 0$$

$$\Rightarrow m = 2, -3$$

$$\text{C.F.} = \phi_1(y+2x) + \phi_2(y-3x)$$

$$\text{P.I.} = \frac{1}{D^2 + DD' - 6D'^2} \cdot y \cos x$$

$$= \frac{1}{(D-2D')(D+3D')} \cdot y \cos x$$

Putting, $y - 3x = C_1 \Rightarrow y = C_1 + 3x$

$$\text{P.I.} = \frac{1}{(D-2D')} \int \underbrace{(C_1 + 3x)}_{\text{I}} \underbrace{\cos x}_{\text{II}} dx$$

$$= \frac{1}{(D-2D')} \left\{ (C_1 + 3x) \sin x - \int 3 \sin x dx \right\}$$

$$= \frac{1}{(D-2D')} \left\{ (C_1 + 3x) \sin x + 3 \cos x \right\}$$

$$P.I. = \frac{1}{(D-2D')} [y \sin x + 3 \cos x]$$

Again, putting, $y+2x = C_2 \Rightarrow y = C_2 - 2x$

$$P.I. = \int \{ (C_2 - 2x) \sin x + 3 \cos x \} dx$$

$$= -(C_2 - 2x) \cos x - \int (-2)(-\cos x) + 3 \sin x$$

$$= -(C_2 - 2x) \cos x - 2 \sin x + 3 \sin x$$

$$= -(C_2 - 2x) \cos x + \sin x$$

$$= -y \cos x + \sin x$$

Hence, the solution of given P.D.E. is,

$$z = \Phi_1(y+2x) + \Phi_2(y-3x) - y \cos x + \sin x.$$

Non-homogeneous Linear Partial Differential Equation with Constant Coefficient

If the order of all the partial derivative involved in the equation are not equal then such ~~homogeneous~~ partial d.e. is known as non-homogeneous linear P.D.E. with constant coefficient.

Example:

$$\frac{\partial^2 z}{\partial x^2} - \frac{\partial^3 z}{\partial y^3} + \frac{\partial z}{\partial x} = x + y.$$

Questions

1. Solve: $(D^2 - D'^2 + D - D')z = 0$

Solⁿ:

The given P.D.E. is,

$$(D^2 - D'^2 + D - D')z = 0$$

$$\Rightarrow [(D - D')(D + D') + (D - D')]z = 0$$

$$\Rightarrow \{(D - D')(D + D' + 1)\}z = 0$$

$$\text{C.F.} = \phi_1(y+x) + e^{-x} \phi_2(y-x)$$

$$\text{P.I.} = 0$$

$\therefore$ The solution of given P.D.E. is,

$$z = \phi_1(y+x) + e^{-x} \phi_2(y-x).$$

2. Solve: $DD'(D-2D'-3)z=0$

Solⁿ:

The given P.D.E. is,
 $DD'(D-2D'-3)z=0$

$$\Rightarrow D=0, D'=0, (D-2D'-3)=0$$

$$C.F. = \phi_1(y) + \phi_2(x) + e^{3x}\phi_3(y+2x)$$

Hence, the solution is,

$$z = \phi_1(y) + \phi_2(x) + e^{3x}\phi_3(y+2x)$$

3. Solve: $(2D-3)(3D-5D'-7)^2z=0$

Solⁿ:

The given P.D.E. is,
 $(2D-3)(3D-5D'-7)^2z=0$

$$C.F. = e^{3x}\phi_1(2y) + e^{7x}\phi_2(3y+5x) + e^{7x}x\phi_3(3y+5x)$$

$$\therefore z = C.F.$$

6. Solve: $(D+2D'-3)(D^2+D')z=0$

Solⁿ:

The given eqⁿ. is,

$$(D+2D'-3)(D^2+D')z=0$$

The C.F. of $(D+2D'-3)$ is $e^{3x} \phi(y-2x)$

For (D^2+D') , let $z = Ae^{hx+ky}$ be a trial solution.

Then,

$$D_z^2 z = Ah^2 e^{hx+ky}$$

$$D'_1 z = Ak e^{hx+ky}$$

$$\text{So, } Ah^2 e^{hx+ky} + Ak e^{hx+ky} = 0$$

$$\Rightarrow A(h^2 + K) e^{hx+ky} = 0$$

$$\Rightarrow h^2 + K = 0$$

$$\Rightarrow h^2 = -K$$

$$\Rightarrow h = \pm \sqrt{-K}$$

$$\Rightarrow h = \pm i\sqrt{K}$$

$\therefore$ C.F. of (D^2+D') is $\sum A e^{\pm i\sqrt{K}x + Ky}$
where, A & K are arbitrary constant.

Hence, the solution of given P.D.E. is,

$$z = e^{3x} \phi(y-2x) + \sum A e^{\pm i\sqrt{K}x + Ky}$$

7. Solve: $(D^2 - D'^2 + D - D')z = e^{2x+3y}$

Solⁿ:

The given eqⁿ is,

$$(D^2 - D'^2 + D - D')z = e^{2x+3y}$$

$$\Rightarrow [(D - D')(D + D') + (D - D')]z = e^{2x+3y}$$

$$\Rightarrow (D - D')(D + D' + 1)z = e^{2x+3y}$$

$$C.F. = \phi_1(y+x) + e^{-x} \phi_2(y-x)$$

$$P.I. = \frac{1}{D^2 - D'^2 + D - D'} \cdot e^{2x+3y}$$

$$\text{let, } v = 2x + 3y$$

$$\Rightarrow D = 2 \quad \& \quad D' = 3$$

$$P.I. = \frac{1}{2^2 - 3^2 + 2 - 3} \int e^v dv$$

$$= -\frac{1}{6} e^v$$

$$= -\frac{1}{6} e^{2x+3y}$$

$$\therefore z = C.F. + P.I. = \phi_1(y+x) + e^{-x} \phi_2(y-x) - \frac{1}{6} e^{2x+3y}$$

8. Solve: $(D^2 + DD' + D' - 1)z = \sin(x + 2y)$

Solⁿ:

The given P.D.E. is,

$$(D^2 + DD' + D' - 1)z = \sin(x + 2y)$$

$$\Rightarrow [(D^2 - 1) + D'(D + 1)]z = \sin(x + 2y)$$

$$\Rightarrow [(D + 1)(D - 1) + D'(D + 1)]z = \sin(x + 2y)$$

$$\Rightarrow (D + 1)(D - 1 + D')z = \sin(x + 2y)$$

$$\Rightarrow (D + 1)(D + D' - 1)z = \sin(x + 2y)$$

$$C.F. = e^{-x} \phi_1(y) + e^x \phi_2(y - x)$$

$$P.I. = \frac{1}{(D^2 + DD' + D' - 1)} \cdot \sin(x + 2y)$$

• Compare with $ax + by$, $a = 1$, $b = 2$
 $\Rightarrow D^2 = -1^2$

$$D \cdot D' = -a \cdot b = -1 \cdot 2 = -2$$

$$D'^2 = -2^2 = -4$$

$$P.I. = \frac{1}{1^2 + (-2) + (-4) + D' - 1} \cdot \sin(x + 2y)$$

$$= \frac{1}{(D' - 4)} \sin(x + 2y)$$

$$= \left(\frac{D' + 4}{D'^2 - 16} \right) \sin(x + 2y)$$

$$= \frac{(D'+4) \sin(x+2y)}{-2^2-16}$$

$$= -\frac{1}{20} (D'+4) \sin(x+2y)$$

$$= -\frac{1}{20} [2 \cos(x+2y) + 4 \sin(x+2y)]$$

$$= -\frac{1}{10} [\cos(x+2y) + 2 \sin(x+2y)]$$

$$\therefore z = e^{-x} \phi_1(y) + e^x \phi_2(y-x) - \frac{1}{10} \cos(x+2y) - \frac{1}{5} \sin(x+2y).$$

9. Solve: $(D - D'^2)z = \cos(3x - y)$

Solⁿ:

The given P.D.E is,
 $(D - D'^2)z = \cos(3x - y)$

Let, $z = Ae^{hx+ky}$ be the C.F. of given P.D.E.

Then, $Dz = Ahe^{hx+ky}$
 $D'^2z = Ak^2e^{hx+ky}$

So, $Ahe^{hx+ky} - Ak^2e^{hx+ky} = 0$

$$\Rightarrow A(h - k^2)e^{hx+ky} = 0$$

$$\Rightarrow h - k^2 = 0$$

$$\Rightarrow h = k^2$$

$$\therefore \text{C.F.} = \sum Ae^{k^2x+ky}$$

where, A & k are arbitrary constants.

$$\text{P.I} = \frac{1}{(D - D'^2)} \cdot \cos(3x - y)$$

Comparing $(3x - y)$ with $(ax + by)$, we get,

$$a = 3 \quad \& \quad b = -1$$

$$D^2 = -a^2 = -3^2 = -9$$

$$D.D' = -ab = 3$$

$$D'^2 = -b^2 = -(-1)^2 = -1$$

$$P.I. = \frac{1}{(D - (-1))} \cdot \cos(3x - y)$$

$$= \frac{1}{(D + 1)} \cdot \cos(3x - y)$$

$$= \frac{(D - 1)}{(D + 1)(D - 1)} \cdot \cos(3x - y)$$

$$= \left(\frac{D - 1}{D^2 - 1} \right) \cos(3x - y)$$

$$= \frac{(D - 1)}{-9 - 1} \cdot \cos(3x - y)$$

$$= -\frac{1}{10} [D \cos(3x - y) - \cos(3x - y)]$$

$$= -\frac{1}{10} [-\sin(3x - y) \times 3 - \cos(3x - y)]$$

$$= \frac{3}{10} \left[\sin(3x - y) + \frac{\cos(3x - y)}{3} \right]$$

Hence, the solution of given P.D.E. is,

$$z = \sum A e^{K^2 x + Ky} + \frac{3}{10} \sin(3x - y) + \frac{1}{10} \cos(3x - y).$$

10. Solve: $(D-1)(D-D'+1)z = 1+xy$

Solⁿ:

The given P.D.E. is,

$$(D-1)(D-D'+1)z = 1+xy$$

$$C.F. = e^x \phi_1(y) + e^{-x} \phi_2(y+x)$$

$$P.I. = \frac{1+xy}{(D-1)(D-D'+1)}$$

$$= -(D-1)^{-1} \{1+(D-D')\}^{-1} \{1+xy\}$$

$$= -(1+D+\dots)(1-D+D'-2DD'+\dots)(1+xy)$$

$$= -(1+D'-DD'+\dots)(1+xy)$$

$$= -(1+xy+x-1)$$

$$= -xy - x$$

$$\therefore z = e^x \phi_1(y) + e^{-x} \phi_2(y+x) - xy - x.$$

Classification of P.D.E.

Consider a second order P.D.E.,

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + F(x, y, u, p, q) = 0 \quad \text{--- (1)}$$

where, A, B, C are constants or continuous function of x & y .

The eqⁿ. (1) is,

- (i) Parabolic if $B^2 - 4AC = 0$
- (ii) Elliptic if $B^2 - 4AC < 0$
- (iii) Hyperbolic if $B^2 - 4AC > 0$

Questions

Q. Classify the following equations.

$$(i) \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 u}{\partial t^2} = 0$$

Solⁿ:

Here, $A=1, B=1$ & $C=1$.

$$B^2 - 4AC = 1^2 - 4 \times 1 \times 1 = -3$$

$\therefore B^2 - 4AC < 0$, therefore, the given is elliptic.

(ii) $4 \frac{\partial^2 u}{\partial x^2} + 4 \frac{\partial u}{\partial x \partial t} + \frac{\partial^2 u}{\partial t^2} = 0$

Solⁿ:

Here, $A=4$, $B=4$ & $C=1$

$$B^2 - 4AC = 4^2 - 4 \times 4 \times 1 = 0$$

$$\therefore B^2 - 4AC = 0$$

$\therefore$ The eqⁿ. is parabolic.

(iii) $x^2 \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = u$

Solⁿ:

Here, $A=x^2$, $B=0$ & $C=-1$

$$\text{Now, } B^2 - 4AC = 0^2 - 4 \times x^2 \times (-1) = 4x^2$$

If $x=0$, then $B^2 - 4AC = 0$, then the given eqⁿ. is parabolic.

If $x \neq 0$, then $B^2 - 4AC > 0$, then the given eqⁿ. is hyperbolic.

$$(iv) \quad x^2 \frac{\partial^2 u}{\partial x^2} + t \frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 u}{\partial t^2} = 0$$

Solⁿ:

Here, $A = x^2$, $B = t$ & $C = 1$

$$\text{Now, } B^2 - 4AC = t^2 - 4x^2$$

i.e. if $t^2 = 4x^2$

If $x = 0$ & $t = 0$, then $B^2 - 4AC = 0$, then the eqⁿ. is parabolic.

If $t^2 < 4x^2$, then $B^2 - 4AC < 0$, then the eqⁿ. is elliptic.

If $t^2 > 4x^2$, then $B^2 - 4AC > 0$, then the eqⁿ. is hyperbolic.

Method of Separation of Variables

In this method, we assume that the solution is the product of two functions, each of which involves only one of the variable.

Questions

1. Solve: $\frac{\partial u}{\partial x} = 3 \frac{\partial u}{\partial t}$ using method of separation of variables.

Solⁿ:

The given eqⁿ. is,
$$\frac{\partial u}{\partial x} = 3 \frac{\partial u}{\partial t} \quad \text{--- (1)}$$

be the solution of eqⁿ. (1)

Let, $u = U = XT$ --- (2), where, $X = X(x)$
 $T = T(t)$

Now, differentiating eqⁿ. (2) partially w.r. to x ,
$$\frac{\partial U}{\partial x} = T \frac{\partial X}{\partial x}$$

Again, differentiating eqⁿ. (2) partially w.r. to t ,
$$\frac{\partial U}{\partial t} = X \frac{\partial T}{\partial t}$$

Now, putting these in eqⁿ. (1),

$$T \frac{\partial X}{\partial x} = 3 \left(X \frac{\partial T}{\partial t} \right)$$

$$\Rightarrow T \frac{\partial X}{\partial x} - 3X \frac{\partial T}{\partial t} = 0$$

Dividing the above eqⁿ. by XT ,

$$\frac{1}{X} \frac{\partial X}{\partial x} - \frac{3}{T} \frac{\partial T}{\partial t} = 0$$

$$\Rightarrow \frac{1}{X} \frac{\partial X}{\partial x} = \frac{3}{T} \frac{\partial T}{\partial t} = K \text{ (say)}$$

$$\text{Now, } \frac{1}{X} \frac{\partial X}{\partial x} = K$$

$$\Rightarrow \frac{1}{X} \partial X = K \partial x$$

$$\Rightarrow \int \frac{1}{X} dX = \int K \partial x$$

$$\Rightarrow \log X = Kx + C_1$$

$$\Rightarrow X = e^{Kx+C_1} \quad \text{--- (3)}$$

$$\text{Similarly, } \frac{3}{T} \frac{\partial T}{\partial t} = K$$

$$\Rightarrow \frac{3}{T} \partial T = K \partial t$$

$$\Rightarrow \int \frac{3}{T} dT = \int k dt$$

$$\Rightarrow 3 \log T = kt + C_2$$

$$\Rightarrow T = e^{\frac{kt+C_2}{3}} \quad \text{--- (4)}$$

Now, eqⁿ. (2) becomes,

$$u = e^{kx+C_1} \times e^{\frac{kt+C_2}{3}}$$

$$\Rightarrow u = e^{kx+kt/3} \cdot e^{C_1+C_2/3}$$

$$\Rightarrow u = Be^{kx+kt/3} \quad [B = e^{C_1+C_2/3} = e^{C_1+C_2}]$$

This is the solution of given P.D.E.

2. Use the method of separation of variables to solve the equation $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$,
given that $u(x, 0) = 6e^{-3x}$.

Solⁿ:

The given P.D.E. is,

$$\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u \quad \text{--- (1)}$$

(2)

Let, $u = XT$ be the solution of above equation.

Now, differentiating eqⁿ. (2) w.r.to x , we get,

$$\frac{\partial u}{\partial x} = T \frac{\partial X}{\partial x}$$

Again, differentiating eqⁿ. (2) w.r.to t , we get,

$$\frac{\partial u}{\partial t} = X \frac{\partial T}{\partial t}$$

Now, putting the value of above two in eqⁿ. (1),

$$T \frac{\partial X}{\partial x} = 2X \frac{\partial T}{\partial t} + XT$$

Dividing above eqⁿ. by XT ,

$$\frac{1}{X} \cdot \frac{\partial X}{\partial x} = \frac{2}{T} \cdot \frac{\partial T}{\partial t} + 1 = K \text{ (Say)}$$

$$\text{Then, } \frac{1}{X} \frac{\partial X}{\partial x} = K$$

$$\Rightarrow \frac{\partial X}{\partial x} = KX$$

$$\Rightarrow \left(\frac{\partial}{\partial x} - K \right) X = 0$$

$$\Rightarrow (D - K)X = 0 \quad \left[\frac{\partial}{\partial x} \equiv D \right]$$

The A.E. is,

$$D - K = 0$$

$$\Rightarrow D = K$$

$$\therefore X = C_1 e^{Kx}$$

$$\text{Also, } \frac{2}{T} \cdot \frac{\partial T}{\partial t} + 1 = K$$

$$\Rightarrow 2 \frac{\partial T}{\partial t} + T = KT$$

$$\Rightarrow \frac{\partial T}{\partial t} + \frac{T}{2} - \frac{KT}{2} = 0$$

$$\Rightarrow [D' + 1/2 - K/2]T = 0$$

$$\Rightarrow D' - (K/2 - 1/2) = 0 \quad [\text{A.E.}]$$

$$\Rightarrow D' = K/2 - 1/2$$

$$\therefore T = C_2 e^{(K/2 - 1/2)t}$$

Putting the values of X & T in eqⁿ. (2),

$$u = C_1 e^{kx} \times C_2 e^{(\frac{k-1}{2})t}$$

$$\Rightarrow u = C e^{kx} \cdot e^{(\frac{k-1}{2})t} \quad \text{--- (3)} [C = C_1 \times C_2]$$

Given, $u(x, 0) = 6 e^{-3x}$

$$\Rightarrow C e^{kx} \cdot e^{(\frac{k-1}{2}) \times 0} = 6 e^{-3x}$$

$$\Rightarrow C e^{kx} = 6 e^{-3x}$$

Comparing, we get,

$$C = 6 \quad \& \quad k = -3$$

Putting in eqⁿ. (3),

$$u = 6 e^{-3x} \cdot e^{(\frac{-3-1}{2})t}$$

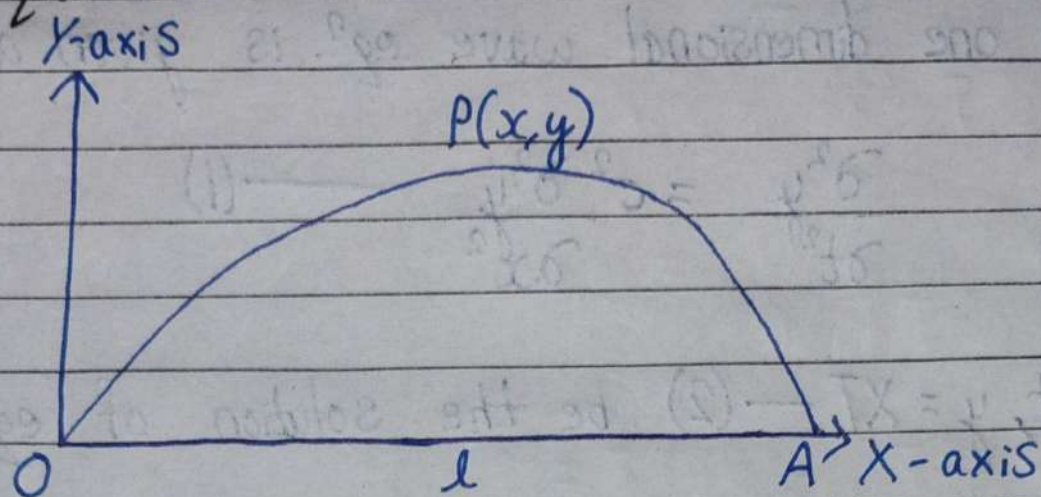
$$\Rightarrow u = 6 e^{-3x} \cdot e^{-2t}$$

$$\therefore u = 6 e^{-(3x+2t)}$$

This is the required solution of given P.D.E.

Equation of Vibrating String (One Dimension)

Wave Equation



Consider a uniform elastic string of length ' l ' stretched tightly between two points O & A and displaced slightly from its equilibrium position OA. Taking the end O as the X-axis and a perpendicular line through O as the Y-axis.

When the string is in motion in the XY-plane, the displacement y of any point of the string is a function of x & t .

Now, the equation of motion for the string is:-

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad [\text{where, } c \text{ is constant}]$$

which is known as the one dimensional wave equation because the motion is only in vertical direction.

Imp. Solution of Wave Equation

The one dimensional wave eqⁿ. is given as,

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{--- (1)}$$

Let, $y = XT$ --- (2) be the solution of eqⁿ. (1).

Differentiating eqⁿ. (2) w.r.to x ,

$$\frac{\partial^2 y}{\partial x^2} = T \frac{\partial^2 X}{\partial x^2}$$

Again, differentiating eqⁿ. (2) w.r.to t ,

$$\frac{\partial^2 y}{\partial t^2} = X \frac{\partial^2 T}{\partial t^2}$$

Putting these two eqⁿ. in eqⁿ. (1),

$$X \frac{\partial^2 T}{\partial t^2} = c^2 T \frac{\partial^2 X}{\partial x^2}$$

Dividing above eqⁿ. by XT ,

$$\frac{1}{T} \frac{\partial^2 T}{\partial t^2} = \frac{c^2}{X} \frac{\partial^2 X}{\partial x^2} = K \text{ (say)}$$

Case - I: When $k=0$.

$$\frac{1}{c^2 T} \frac{\partial^2 T}{\partial t^2} = 0$$

$$\Rightarrow \frac{\partial^2 T}{\partial t^2} = 0$$

$$\Rightarrow \iint \frac{\partial^2 T}{\partial t^2} = 0$$

$$\Rightarrow \int \frac{\partial T}{\partial t} = C_1$$

$$\Rightarrow T = C_1 t + C_2$$

$$\frac{1}{X} \frac{\partial^2 X}{\partial x^2} = 0$$

$$\Rightarrow \frac{\partial^2 X}{\partial x^2} = 0$$

$$\Rightarrow \iint \frac{\partial^2 X}{\partial x^2} = 0$$

$$\Rightarrow \int \frac{\partial X}{\partial x} = C_3$$

$$\Rightarrow X = C_3 x + C_4$$

Putting these values in eqⁿ. (2),

$$y = (C_1 t + C_2)(C_3 x + C_4)$$

Since, the wave equation is a periodic function of x & t . Therefore, it must contain trigonometrical term. So, from above it is concluded that the solution is not the solution of eqⁿ. (1).

Case - II: When $k > 0$.

$$\frac{1}{c^2 T} \frac{\partial^2 T}{\partial t^2} = \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = \rho^2$$

$$\Rightarrow \frac{1}{c^2 T} \frac{\partial^2 T}{\partial t^2} = \rho^2$$

$$\Rightarrow \frac{\partial^2 T}{\partial t^2} - \rho^2 c^2 T = 0$$

$$\Rightarrow (D^2 - \rho^2 c^2) T = 0$$

$$\Rightarrow D^2 - \rho^2 c^2 = 0 \quad [\text{A.E.}]$$

$$\Rightarrow D = \pm \rho c$$

$$\therefore T_1 = C_1 e^{\rho c t} + C_2 e^{-\rho c t}$$

$$\text{Also, } \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = \rho^2$$

$$\Rightarrow \frac{\partial^2 X}{\partial x^2} = \rho^2 X$$

$$\Rightarrow D^2 X - \rho^2 X = 0$$

$$\Rightarrow (D^2 - \rho^2) X = 0$$

$$\Rightarrow D^2 - \rho^2 = 0 \quad [\text{A.E.}]$$

$$\Rightarrow D = \pm p$$

$$\therefore X = C_3 e^{px} + C_4 e^{-px}$$

Putting X & T in eqⁿ. (2),

$$y = (C_1 e^{pct} + C_2 e^{-pct}) (C_3 e^{px} + C_4 e^{-px})$$

This is also not the solution.

Case - III: When $k < 0$.

$$\frac{1}{C^2 T} \frac{\partial^2 T}{\partial t^2} = \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = -p^2$$

Taking,

$$\frac{1}{C^2 T} \frac{\partial^2 T}{\partial t^2} = -p^2$$

$$\Rightarrow \frac{\partial^2 T}{\partial t^2} + p^2 C^2 T = 0$$

$$\Rightarrow (D^2 + p^2 C^2) T = 0$$

The A.E. is,

$$D^2 + p^2 C^2 = 0$$

$$\Rightarrow D = \pm i p C$$

$$T = C_1'' \cos p C t + C_2'' \sin p C t$$

$$\text{Also, } \frac{1}{X} \frac{\partial^2 X}{\partial x^2} = -p^2$$

$$\Rightarrow \frac{\partial^2 X}{\partial x^2} + p^2 X = 0$$

$$\Rightarrow (D^2 + p^2)X = 0$$

The A.E. is,

$$D^2 + p^2 = 0$$

$$\Rightarrow D = \pm ip$$

$$X = C_3'' \cos px + C_4'' \sin px$$

Putting the values of X & T in eqⁿ. (2),

$$y = (C_1'' \cos pCt + C_2'' \sin pCt)(C_3'' \cos px + C_4'' \sin px)$$

This^{is} the solution of the wave equation.

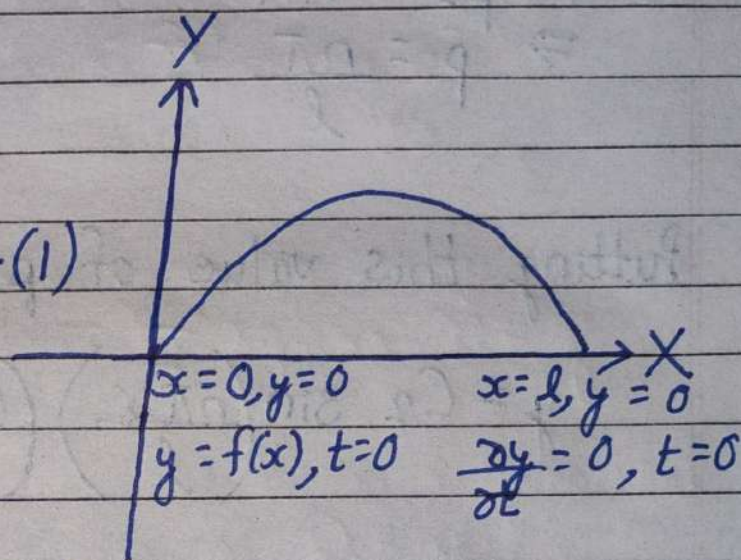
Q.1) Solve completely the equation $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$

representing the vibrations of a string of length l , fixed at both ends, given that $y(0, t) = 0$, $y(l, t) = 0$, $y(x, 0) = f(x)$ and $\frac{\partial y}{\partial t}(x, 0) = 0$, $0 < x < l$.

Solⁿ:

Given P.D.E. is,

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{--- (1)}$$



The solution of eqⁿ. (1) is,

$$y = (C_1 \cos px + C_2 \sin px) (C_3 \cos Cpt + C_4 \sin Cpt) \quad \text{--- (2)}$$

Now, applying the boundary condition $x=0$ & $y=0$ in eqⁿ. (2),

$$0 = (C_1 + 0) (C_3 + 0)$$

$$0 = C_1 (C_3 \cos Cpt + C_4 \sin Cpt)$$

$$\Rightarrow C_1 = 0$$

Putting this value of C_1 in eqⁿ. (2),

$$y = C_2 \sin px (C_3 \cos Cpt + C_4 \sin Cpt) \quad \text{--- (3)}$$

Again, applying $x=l$ & $y=0$ in eqⁿ. (3),

$$0 = C_2 \sin pl (C_3 \cos Cpt + C_4 \sin Cpt)$$

$$\Rightarrow \sin pl = 0$$

$$\Rightarrow \sin pl = \sin n\pi$$

$$\Rightarrow pl = n\pi$$

$$\Rightarrow p = \frac{n\pi}{l}$$

Putting this value of p in eqⁿ. (3),

$$y = C_2 \sin\left(\frac{n\pi x}{l}\right) \left(C_3 \cos \frac{n\pi Ct}{l} + C_4 \sin \frac{n\pi Ct}{l} \right) \quad \text{--- (4)}$$

Differentiating eqⁿ. (4) w.r.to t , we get,

$$\frac{\partial y}{\partial t} = C_2 \sin\left(\frac{n\pi x}{l}\right) \left[-\frac{n\pi C}{l} \cdot C_3 \sin \frac{n\pi Ct}{l} + \frac{n\pi C}{l} \cdot C_4 \cos \frac{n\pi Ct}{l} \right]$$

Applying the condition $\frac{\partial y}{\partial t} = 0$ & $t=0$ in above eqⁿ,

$$0 = C_2 \sin\left(\frac{n\pi x}{l}\right) \left[0 + \frac{n\pi C}{l} \cdot C_4 \right]$$

$$\Rightarrow C_4 = 0$$

Putting this value of C_4 in eqⁿ. (4), we get,

$$y = C_2 \sin\left(\frac{n\pi x}{l}\right) \left[C_3 \cos\left(\frac{n\pi Ct}{l}\right) + 0 \right]$$
$$\Rightarrow y = C_2 \cdot C_3 \cdot \sin\left(\frac{n\pi x}{l}\right) \cdot \cos\left(\frac{n\pi Ct}{l}\right) \text{ --- (5)}$$

$$\Rightarrow y = b_n \sin\left(\frac{n\pi x}{l}\right) \cdot \cos\left(\frac{n\pi Ct}{l}\right)$$

[where $C_2 \cdot C_3 = b_n$]

The most general solution is,

$$y = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \cdot \cos\left(\frac{n\pi Ct}{l}\right) \text{ --- (6)}$$

Now, applying $y = f(x)$ & $t = 0$ in eqⁿ. (6),

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$\Rightarrow b_n = \frac{2}{l} \int_0^l f(x) \cdot \sin\left(\frac{n\pi x}{l}\right) dx \text{ --- (7)}$$

[Using Fourier Sine Series]

Hence, the required solution is eqⁿ. (6) and b_n is in eqⁿ. (7).

Q.2) If a string of length l is initially at rest in equilibrium position and each of its points is given that velocity,

$$\left(\frac{\partial y}{\partial t}\right)_{t=0} = b \sin^3\left(\frac{\pi x}{l}\right)$$

Solⁿ:

The one dimension wave eqⁿ. is,

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \text{--- (1)}$$

The solution of eqⁿ. (1) is,

$$y = (C_1 \cos px + C_2 \sin px) (C_3 \cos cpt + C_4 \sin cpt) \quad \text{--- (2)}$$

Boundary conditions,

$$y = 0 \text{ \& } x = 0 \text{ --- (i)}$$

$$y = 0 \text{ \& } x = l \text{ --- (ii)}$$

$$y = 0 \text{ \& } t = 0 \text{ --- (iii)}$$

$$\left(\frac{\partial y}{\partial t}\right) = b \sin^3\left(\frac{\pi x}{l}\right) \text{ \& } t = 0 \text{ --- (iv)}$$

- Applying condition $y = 0$ \& $x = 0$ in eqⁿ. (2),

$$0 = C_1 (C_3 \cos cpt + C_4 \sin cpt)$$

$$\Rightarrow C_1 = 0$$

Putting this value of C_1 in eqⁿ. (2),

$$y = C_2 \sin px (C_3 \cos cpt + C_4 \sin cpt) \text{ --- (3)}$$

- Applying condition $y=0$ & $x=l$ in eqⁿ. (3),

$$0 = C_2 \sin pl (C_3 \cos cpt + C_4 \sin cpt)$$

$$\Rightarrow C_2 \sin pl = 0$$

$$\Rightarrow \sin pl = 0$$

$$\Rightarrow \sin pl = \sin n\pi$$

$$\Rightarrow pl = n\pi$$

$$\Rightarrow p = \frac{n\pi}{l}$$

Putting this value of p in eqⁿ. (3),

$$y = C_2 \sin\left(\frac{n\pi x}{l}\right) \left[C_3 \cos\left(\frac{n\pi ct}{l}\right) + C_4 \sin\left(\frac{n\pi ct}{l}\right) \right] \text{---(4)}$$

Applying the condition $y=0$ & $t=0$ in eqⁿ. (4),

$$0 = C_2 \sin\left(\frac{n\pi x}{l}\right) [C_3 + 0]$$

$$\Rightarrow C_3 = 0$$

Putting this value of C_3 in eqⁿ. (4),

$$y = C_2 \sin\left(\frac{n\pi x}{l}\right) \left[0 + C_4 \sin\left(\frac{n\pi ct}{l}\right) \right]$$

$$\Rightarrow y = C_2 \cdot C_4 \cdot \sin\left(\frac{n\pi x}{l}\right) \cdot \sin\left(\frac{n\pi ct}{l}\right) \text{---(5)}$$

$$\Rightarrow y(x, t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \cdot \sin\left(\frac{n\pi ct}{l}\right) \quad \text{--- (6)}$$

Differentiating eqⁿ. (6) w.r.to t ,

$$\frac{\partial y}{\partial t} = \sum_{n=1}^{\infty} b_n \cdot \left(\frac{n\pi c}{l}\right) \sin\left(\frac{n\pi x}{l}\right) \cdot \cos\left(\frac{n\pi ct}{l}\right)$$

Applying condition $\frac{\partial y}{\partial t} = b \sin^3\left(\frac{\pi x}{l}\right)$ & $t=0$, in above eqⁿ;

$$b \sin^3\left(\frac{\pi x}{l}\right) = \sum_{n=1}^{\infty} b_n \left(\frac{n\pi c}{l}\right) \cdot \sin\left(\frac{n\pi x}{l}\right) \cdot 1$$

$$\Rightarrow \frac{b}{4} \left[3 \sin\left(\frac{\pi x}{l}\right) - \sin\left(\frac{3\pi x}{l}\right) \right] = \sum_{n=1}^{\infty} b_n \left(\frac{n\pi c}{l}\right) \cdot \sin\left(\frac{n\pi x}{l}\right)$$

$$\left[\begin{array}{l} \text{Using, } \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta \\ \Rightarrow \sin^3 \theta = \frac{1}{4} (3 \sin \theta - \sin 3\theta) \end{array} \right]$$

$$\begin{aligned} \frac{b}{4} \left[3 \sin\left(\frac{\pi x}{l}\right) - \sin\left(\frac{3\pi x}{l}\right) \right] &= b_1 \left(\frac{\pi c}{l}\right) \cdot \sin\left(\frac{\pi x}{l}\right) \\ &+ b_2 \left(\frac{2\pi c}{l}\right) \cdot \sin\left(\frac{2\pi x}{l}\right) + b_3 \left(\frac{3\pi c}{l}\right) \cdot \sin\left(\frac{3\pi x}{l}\right) \\ &+ \dots \end{aligned}$$

Equating the coefficients of $\sin\left(\frac{\pi x}{l}\right)$, $\sin\left(\frac{2\pi x}{l}\right)$, $\sin\left(\frac{3\pi x}{l}\right)$, ..., we get,

$$\frac{3b}{4} = b_1 \left(\frac{\pi c}{l}\right) \Rightarrow b_1 = \frac{3bl}{4\pi c}$$

$$0 = b_2 \left(\frac{2\pi c}{l}\right) \Rightarrow b_2 = 0$$

$$\frac{-b}{4} = b_3 \left(\frac{3\pi c}{l}\right) \Rightarrow b_3 = \frac{-bl}{12\pi c}$$

Also, $b_4 = b_5 = \dots = 0$

Putting these values in eqⁿ. (6), we get,

$$y = \frac{3bl}{4\pi c} \cdot \sin\left(\frac{\pi ct}{l}\right) \cdot \sin\left(\frac{\pi x}{l}\right) - \frac{bl}{12\pi c} \cdot \sin\left(\frac{3\pi ct}{l}\right) \cdot \sin\left(\frac{3\pi x}{l}\right) \quad \text{--- (7)}$$

Eqⁿ. (7) is the required solution.